Kunjal Purohit

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Education

University of California, Santa Cruz — Santa Cruz, CA

Bachelor of Science in Computer Science and Bachelor of Science in Computer Engineering — *Expected June 2028*

- **GPA: 3.95/4.0**, Dean's Honors List (3x)
- Relevant Coursework: Data Structures & Algorithms, Computer Systems & C, Assembly Language, Discrete Mathematics, Probability & Statistics, Linear Algebra, Multivariable Calculus, Differential Equations

Skills

Languages: Python, TypeScript, JavaScript, C/C++, Java, Swift, SQL (PostgreSQL), HTML/CSS, Assembly
Frameworks & Libraries: Next.js, React, Node.js, FastAPI, Capacitor, Convex, Playwright, Tailwind CSS, OpenCV
Tools & Platforms: Supabase, Neo4j, Clerk, Plaid, Vercel, Codemagic, Unix/Linux
Embedded Systems: STM32, ESP32, CAN Bus, OneWire, Sensor Integration, Signal Filtering
AI: Coding Agents, LLM Integration, Prompt Engineering

Experience

Embedded Software Engineer, Electrical Branch — October 2025 – Present

Formula Slug (FSAE) — *Santa Cruz, CA*

- Built STM32 C/C++ firmware for real-time vehicle data acquisition, from sensor input to CAN bus telemetry
- Built tray-temperature monitoring on 4 DS18B20 sensors, triggering **automatic shutdown** on safety faults
- Replaced a non-differentiable piecewise torque map with a cubic-polynomial curve (41-point LUT, linear interpolation), now the **default accelerator mapping** in VCU firmware
- Redesigned the CAN protocol for a new battery and chip architecture, **adopted team-wide** for on-car telemetry

Web Development Lead — August 2026 – Present

Association for Computing Machinery (ACM) — *Santa Cruz, CA*

- Lead 4 developers in redesigning the club website, shipping **5 features** before fall recruitment
- Replaced a buggy, non-responsive site that had no working backend with member login, session handling, and a mobile-responsive layout, deployed on Vercel
- Redesigned the events page and navigation to showcase 15+ events

Projects

UCSC Dining Tracker | App Store | *Next.js, TypeScript, Supabase, Python, Swift* — June 2026 – Present

- Shipped a web and iOS macro tracker with **70+ users** and meal-period food suggestions
- Indexed 1,699 campus food items and **233K+ menu records** across 14 dining halls with Python/Playwright scrapers on GitHub Actions, running daily at a **100% success rate across 470+ runs**
- Stored 90+ days of menu history in a Supabase PostgreSQL database with per-user Row-Level Security (RLS)
- Automated iOS builds, code-signing, and App Store deployments with a Codemagic CI/CD pipeline that runs on every merge, producing 64 builds and **4 App Store releases** (v1.0–v1.3)

Dowser (Oil Well Detection) | Live App | *Next.js, Convex, Capacitor, ESP32, YOLO* — October 2026

- Built in 42 hours with a team of 4 at Cal Hacks 13.0 an app that flagged candidate wells by matching well symbols on old USGS topo maps against **158,000 California state well records**
- Built the Capacitor iOS field app and Convex backend, streaming ESP32 readings live to a map with auto-confirm logic (8 readings $\geq$ 1000 within 10 s), and set up the ESP32 hardware

Heirloom (AI Multi-Generational Wealth Platform) | *Next.js, FastAPI, Neo4j, Plaid, Daytona* — October 2026

- **Won 1st place in the Daytona track** (20 teams) at SF Tech Week by building an AI wealth platform in 12 hours with AI agents, a Neo4j knowledge graph, and Plaid, iterating on finance professionals' input
- Designed AI agents that flagged conflicts between parents' financial decisions and heirs' goals for wealth advisors
- Enforced firm-level, multi-tenant data isolation across all database operations by mapping Clerk JWT claims to PostgreSQL Row-Level Security (RLS) policies

Autonomous Vehicle (UCSD COSMOS) | *Python, OpenCV, ROS, LiDAR, GNSS, Jetson Nano* — July 2024

- Enabled **3 fully autonomous laps** on a 5-person team by writing Python drivers for a LiDAR and GNSS receiver, fusing both into one time-aligned log, and generating synthetic LiDAR logs from a kinematic bicycle model
- Built occupancy-grid mapping and localization, validated in simulation and on real indoor/outdoor courses